

Autonomous Theory Construction Laboratory

Lecture 15: Final Holdout

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2026-12-07–2026-12-13

Overview

The final holdout is evaluated once and followed by a concise error analysis rather than another search cycle.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$R_{final} = R_{RC1} \cup O_h$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$releasepass = registry \wedge legacy \wedge controls \wedge replay$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$branch_{next.parent} = hash(snapshot_{submit})$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Final Holdout Packet
- Release Signature Standard
- RLP-12 closure appendix